



SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL
SCIENCES



Technical Presentation

Architecture of Data Warehouse

CSA1606 - DATA WAREHOUSING AND DATA MINING

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INTRODUCTION

What is the Architecture of a Data Warehouse?

Data Warehouse Architecture is the structure that defines how data is collected, processed, stored, and accessed.

It organizes data from different sources into a central repository using the ETL process.

It helps users analyse historical data and generate reports for decision-making.

Importance

Stores historical data.

Supports business decisions.

Improves reporting and analysis.

Core Concepts

Three-Tier Architecture

A Data Warehouse uses a three-tier architecture to collect, store, and analyze data efficiently.

1. Bottom Tier

Collects data from different sources.

Uses the ETL process (Extract, Transform, Load).

Stores data in the Data Warehouse.

2. Middle Tier

Contains the OLAP Server.

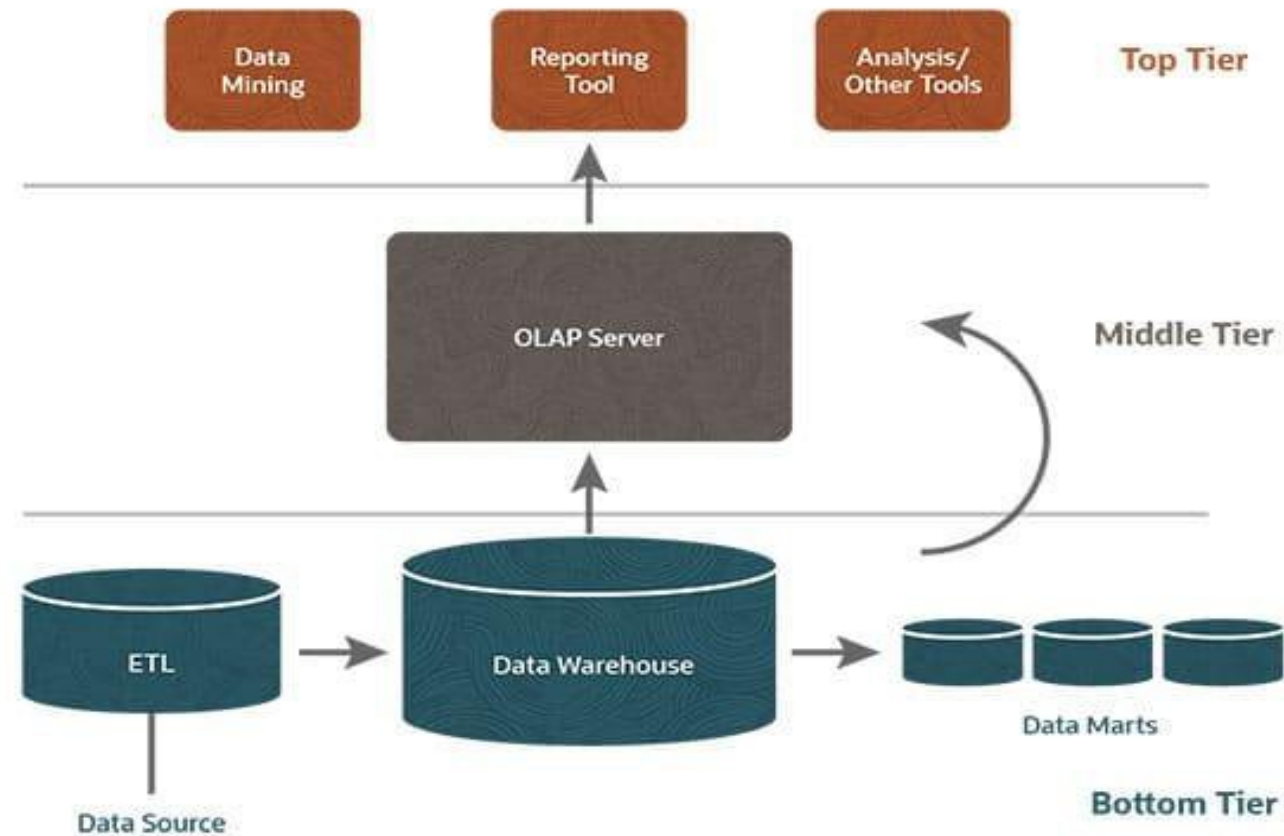
Organizes and processes data for analysis.

3. Top Tier

Used by users to access reports and dashboards.

Includes Business Intelligence (BI) tools for data analysis.

Diagram



ETL PROCESS

ETL (Extract, Transform, Load)

ETL is the process of collecting data from different sources, converting it into a useful format, and storing it in the Data Warehouse.

1. Extract

Collects data from different sources.

Sources include databases, Excel files, websites, and business applications.

2. Transform

Cleans incorrect or duplicate data.

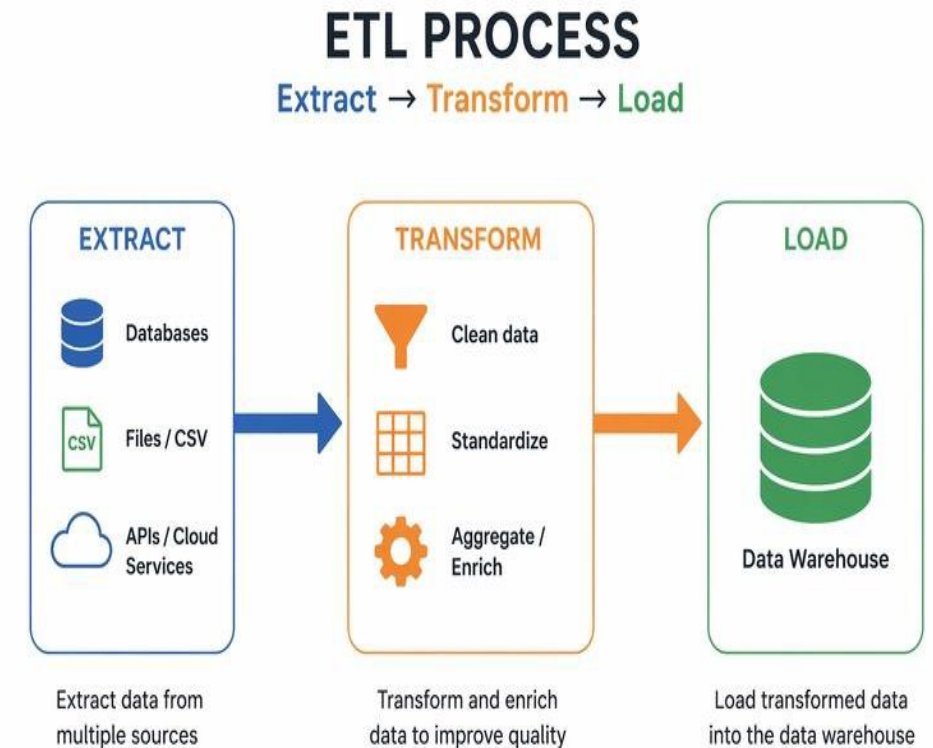
Converts data into a common format.

Improves data quality.

3. Load

Loads the transformed data into the Data Warehouse.

The data is then ready for analysis and reporting.



Data Mart and Metadata

Data Mart

A small part of a Data Warehouse designed for a specific department.

Stores department-specific data for quick analysis.

Examples:

- Sales Data Mart

- Finance Data Mart

- HR Data Mart

Metadata

Metadata means "data about data".

Provides information about data source, format, and location.

Helps users understand and manage stored data.

Metadata Example

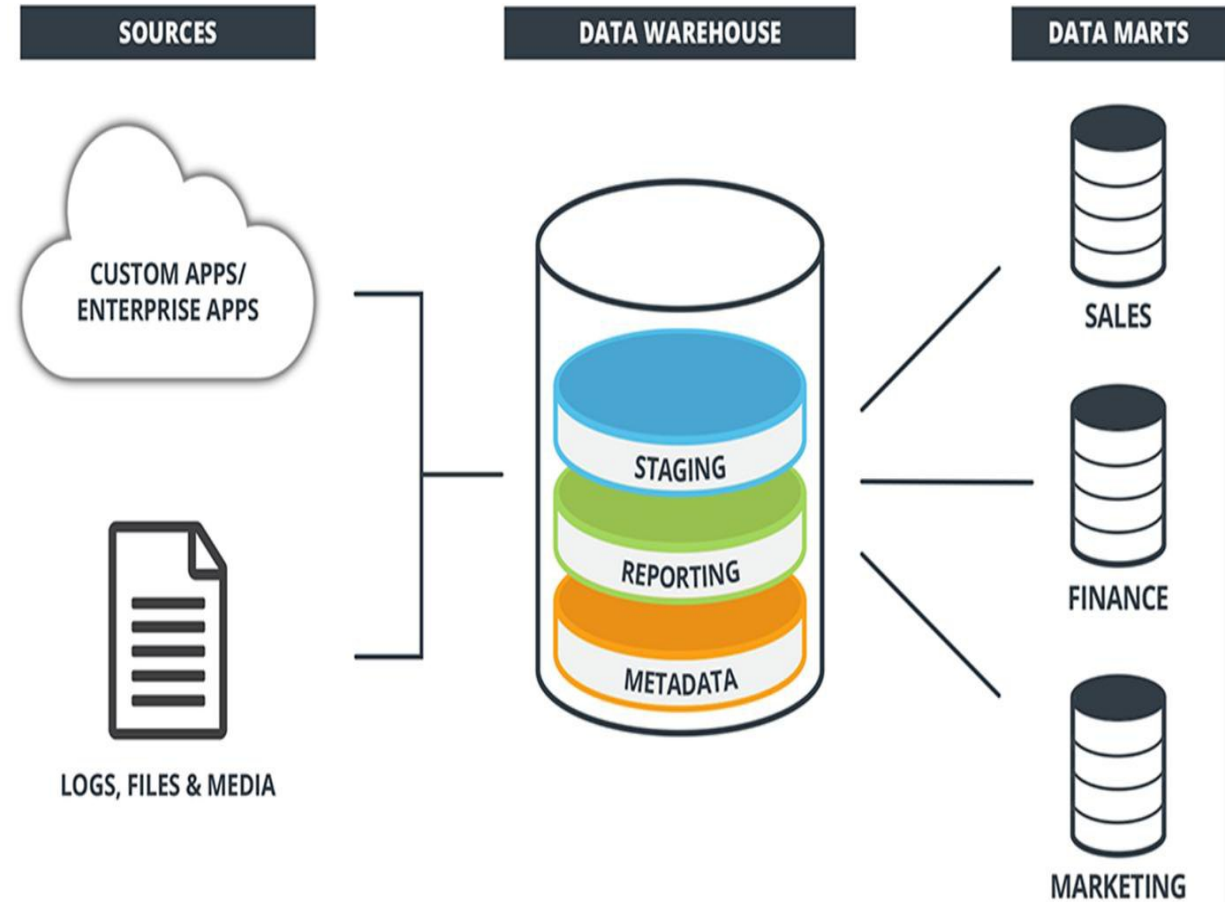
Data: Customer Name, ID, Purchase Details

Metadata:

Data source: Customer Database

Data type: Text/Number

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Working Process of Data Warehouse

Step-by-Step Process

1. Data Collection

Data is collected from different sources like databases, ERP, and CRM.

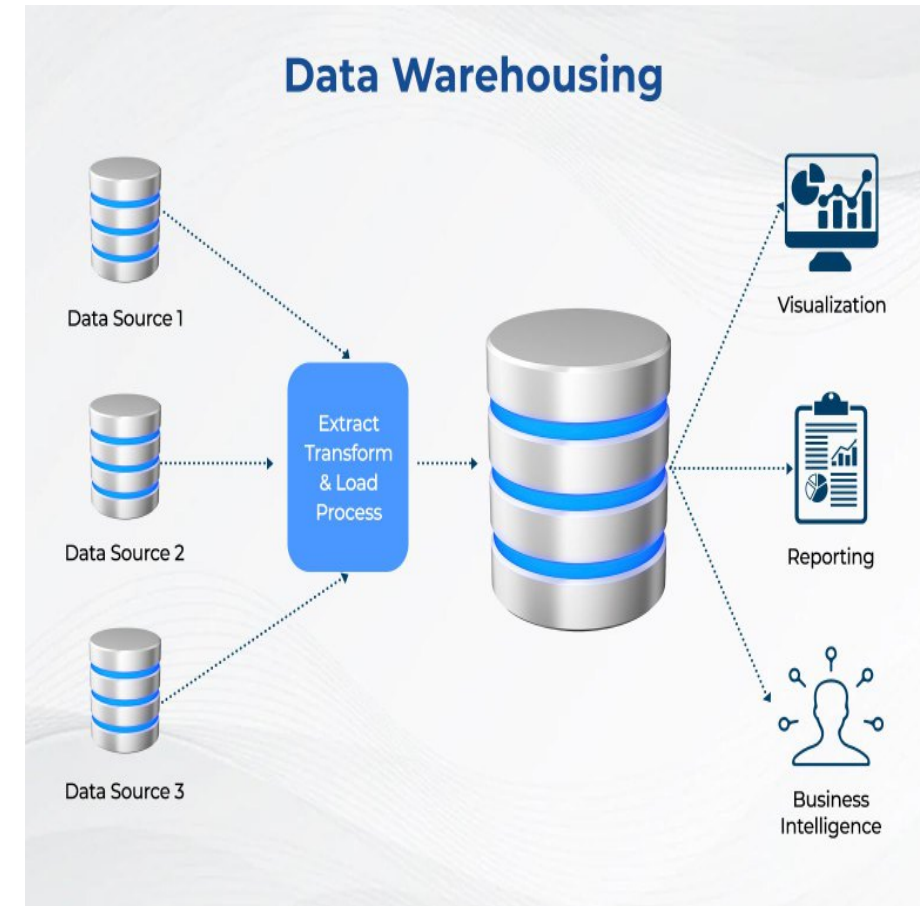
2. ETL Process

Data is extracted, cleaned, and transformed into a proper format.

3. Data Storage

Processed data is stored in the Data Warehouse.

4. Data Analysis



Real-World Applications

A Data Warehouse is used in different industries to store, analyze, and manage large amounts of data for better decision-making.

1. Banking

Analyzes customer transactions.

Detects fraud activities.

Helps in loan and risk analysis.

2. Healthcare

Stores patient records.

Analyzes disease patterns.

Improves hospital management.



3. Retail

Analyses sales data.

Tracks customer behaviour.

Helps in inventory management

4. Education

Analyses student performance.

Maintains academic reports.

Tools Used in Data Warehouse

KNIME – Creates data workflows and performs data analysis.

Power BI – Creates dashboards and visual reports.

Tableau – Provides data visualization and analytics.

PostgreSQL/MySQL – Stores and manages data.

CONCLUSION

A Data Warehouse helps organizations store and analyse large amounts of data from different sources.

It improves decision-making by providing accurate reports and useful insights.

It supports Business Intelligence (BI), analytics, and future technologies like AI and Machine Learning.

Data Warehouses will continue to play an important role in managing and analysing data efficiently.



Engineer to Excel



Thank You